



Asset Performance:

Ene 03 Fabric Performance



Contributes to 40 credits available for asset energy rating



No Minimum Standard

Aim

To minimise operational energy consumption and the associated carbon emissions by increasing the intrinsic energy efficiency of the building fabric and the installed buildings services.

Question

Fabric performance questions:

Credits	Question	Enter value or select a single answer option	Units
-	Average U-value of the external walls		W/m ² K
-	Average U-value of the roof		W/m ² K
-	Average U-value of the windows		W/m ² K
-	External wall area (excluding windows)		m ²
-	Roof area		m ²
-	Window area		m ²
-	Volume of building		m ³
-	Average g-value of the glazing		
-	Is external solar shading to reduce excess solar gains during the cooling season fitted?	Yes/ No	
-	In what year were the windows last replaced?		

Methodology

Asset Energy Calculator Guidance

See Asset Energy Calculator Guidance for the heat loss through fabric and the internal heat gains.

Average U-value and g-value calculation

An area weighted average across all different constructions for the relevant element.

External solar shading

Solar shading should be provided where solar heat gains will be significant during the cooling season. The significance of the solar heat gains is determined by the amount of sunlight entering the room. The Assessor should consider the following characteristics;

- Orientation of the windows to direct sunlight in the cooling season
- Overall size of the windows
- Relative size of the windows compared to the space.

For example, in the northern hemisphere high levels of south facing glazing could result in significant solar heat gains in the summer.

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
All	Evidence demonstrating the U or g values may include: • Energy Performance Certificate • Photographic evidence of building element construction • Building design plans • Written details of the National Building Regulations stating the minimum U-value at the time the asset was constructed • Expert report by a building surveyor or equivalent estimating the U-value • Manufacturer's literature specifying the U-values.
All	Plans and façade drawings to demonstrate calculation of volume of building
All	External solar shading can be demonstrated via photographic evidence or the Assessor's Site Inspection report.

Definitions

U-value:

A measure of heat loss in a building element such as a wall, floor or roof that measures the effectiveness of a material as an insulator. A lower U-value indicates a higher level of thermal efficiency.

g-value:

A measure of the proportion of heat that is transmitted through a window where a g-value of 1 indicates 100% heat transmission.

Additional information

Asset energy calculator

In order to obtain the maximum possible number of credits for the asset energy performance as much information as possible should be entered. Where no value is entered, the asset energy rating calculator will assume a pessimistic default value.